

1. Historical Data Average

This appendix presents historical data about mutation operators for the C and Python languages we collected. These data can serve as baseline values for organizations that do not yet have their own historical data. As with other technical debt tools, such as SonarQube, organizations should calibrate these values over time based on their specific context.

1.1. Historical Data for the C language

Data for the C language correspond to data collected from several experiments, in which we determined all equivalent mutants by hand and reported the average percentage of equivalent mutants for each mutation operator in *Proteum/IM* unit testing tools Delamaro, Maldonado and Vincenzi (2001). We use the historical data from Delamaro, Deng, Durelli, Li and Offutt (2014). From this study, the average time for deciding equivalence or creating a new test case to kill a non-equivalent mutant was 15 minutes, which is also consistent with experiments from other researchers Schuler and Zeller (2013); Guerino, Kuroishi, Paiva and Vincenzi (2024). Indeed, this time can vary from operator to operator, but in this experiment, Delamaro et al. (2014) did not collect data at such fine granularity. Users may collect these average times for equivalence detection and test case creation per mutation operator once the formulas are indexed by the mutation operators' time.

Table 1 presents the percentage of equivalent and non-equivalent mutants for the historical data, considering the subjects in this experiment (see Table ??). They contain all unit mutation operators for the C language implemented in the *Proteum/IM* mutation testing tool. More information about the tool and its set of mutation operators can be found elsewhere Delamaro et al. (2001).

Table 1: Consolidated Mutation Operators Report

Operator	Total	Non-Eq	Eq	%Non-Eq	%Eq
CCCR	4624	4133	491	89.4	10.6
CCDL	457	400	57	87.5	12.5
CCSR	5959	5711	248	95.8	4.2
CRCR	7414	7071	343	95.4	4.6
OAAA	174	173	1	99.4	0.6
OAAN	1188	1149	39	96.7	3.3
OABA	87	86	1	98.9	1.1
OABN	720	662	58	91.9	8.1
OAEA	48	48	0	100.0	0.0
OALN	632	616	16	97.5	2.5
OARN	1896	1852	44	97.7	2.3
OASA	58	58	0	100.0	0.0
OASN	480	460	20	95.8	4.2
OBAA	0	0	0	0.0	0.0
OBAN	60	58	2	96.7	3.3
OBBA	0	0	0	0.0	0.0
OBBN	24	23	1	95.8	4.2
OBEA	0	0	0	0.0	0.0
OBLN	24	23	1	95.8	4.2
OBNG	36	36	0	100.0	0.0
OBRN	72	71	1	98.6	1.4
OBSA	22	22	0	100.0	0.0
OBSN	24	24	0	100.0	0.0
OCNG	426	422	4	99.1	0.9
OCOR	122	29	93	23.8	76.2
OEAA	2300	1944	356	84.5	15.5
OEBA	1278	918	360	71.8	28.2

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Table 1: Consolidated Mutation Report

Operator	Total	Non-Eq	Eq	%Non-Eq	%Eq
OESA	852	751	101	88.1	11.9
OIDO	187	183	4	97.9	2.1
OIPM	2	2	0	100.0	0.0
OLAN	531	404	127	76.1	23.9
OLBN	315	196	119	62.2	37.8
OLLN	110	103	7	93.6	6.4
OLNG	330	324	6	98.2	1.8
OLRN	660	569	91	86.2	13.8
OLSN	210	184	26	87.6	12.4
OODL	1930	1772	158	91.8	8.2
ORAN	2004	1753	251	87.5	12.5
ORBN	1140	970	170	85.1	14.9
ORLN	854	768	86	89.9	10.1
ORRN	2135	1847	288	86.5	13.5
ORSN	760	668	92	87.9	12.1
OSAA	0	0	0	0.0	0.0
OSAN	0	0	0	0.0	0.0
OSBA	0	0	0	0.0	0.0
OSBN	0	0	0	0.0	0.0
OSEA	0	0	0	0.0	0.0
OSLN	0	0	0	0.0	0.0
OSRN	0	0	0	0.0	0.0
OSSA	0	0	0	0.0	0.0
OSSN	0	0	0	0.0	0.0
SBRC	7	4	3	57.1	42.9
SBRn	2	2	0	100.0	0.0
SCRB	2	2	0	100.0	0.0
SCRn	0	0	0	0.0	0.0
SDWD	0	0	0	0.0	0.0
SGLR	10	1	9	10.0	90.0
SMTc	117	108	9	92.3	7.7
SMTT	117	117	0	100.0	0.0
SMVB	69	65	4	94.2	5.8
SRSR	2640	2473	167	93.7	6.3
SSDL	2062	1916	146	92.9	7.1
SSWM	118	113	5	95.8	4.2
STRI	590	569	21	96.4	3.6
STRP	2070	2036	34	98.4	1.6
SWDD	43	28	15	65.1	34.9
VDTR	4278	2946	1332	68.9	31.1
VGAR	145	123	22	84.8	15.2
VGPR	81	65	16	80.2	19.8
VGSR	1447	1348	99	93.2	6.8
VGTR	0	0	0	0.0	0.0
VLAR	48	41	7	85.4	14.6
VLPR	265	262	3	98.9	1.1
VLSR	8257	7791	466	94.4	5.6

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Table 1: Consolidated Mutation Report

Operator	Total	Non-Eq	Eq	%Non-Eq	%Eq
VLTR	0	0	0	0.0	0.0
VSCR	335	324	11	96.7	3.3
VTWD	2852	2585	267	90.6	9.4
VVDL	818	772	46	94.4	5.6
Total	66518	60174	6344	90.5	9.5

Table 1 presents the total number of mutants, number of non-equivalent and equivalent mutants per mutation operator, and the respective percentage of non-equivalent and equivalent mutants with respect to the total per mutation operator. Observe that some mutation operators contain no data since they did not generate mutants for the subject programs.

1.2. Historical Data for the Python language

Although we did not illustrate the calculation of *MTD* for Python, the proposed formulas are independent. Below, we share the historical data we have for the Python programming language, including the mutation operators of two mutation testing tools for Python: MutPy and Cosmic-Ray [Guerino et al. \(2024\)](#).

In the experiment reported by [Guerino et al. \(2024\)](#), we also analyzed equivalent mutants generated by two different mutation testing tools for Python by hand. Table 2 presents the data from MutPy, and Table 3 presents the data from Cosmic-Ray. For more information about the MutPy mutation operators, the interested reader may consult [Hossner, Hałas, Myint and Mueller \(2021\)](#), and for Cosmic-Ray mutation operators, she may consult [Bingham \(2021\)](#).

Tables 2 and 3 present the same structure as Table 1 but considering different sets of mutation operators applied to a set of Python programs described in the work of [Guerino et al. \(2024\)](#).

It is interesting to note that each tool supports different mutation operators, though they may share some. MutPy has a smaller set of mutation operators but generates more mutants (3,091 in total) than Cosmic-Ray (1,948 in total). However, in terms of the percentage of equivalent mutants, mutations from Cosmic-Ray contain 5.6% equivalent mutants. In contrast, mutations from MutPy contain only 2.3% equivalent mutants, considering the full set of mutation operators. The percentages of non-equivalent and equivalent mutants for each mutation operator are important information for estimating *MTD* on any program.

Table 2: MutPy Mutation Operators Report

Operator	Total	Non-Eq	Eq	%Non-Eq	%Eq
AOD	3	3	0	100.0	0.0
AOR	44	43	1	97.7	2.3
ASR	63	63	0	100.0	0.0
BCR	2	2	0	100.0	0.0
CDI	239	237	2	99.2	0.8
COD	19	19	0	100.0	0.0
COI	260	260	0	100.0	0.0
CRP	321	309	12	96.3	3.7
DDL	6	6	0	100.0	0.0
EHD	9	9	0	100.0	0.0
EXS	11	9	2	81.8	18.2
IOD	3	2	1	66.7	33.3
IOP	1	0	1	0.0	100.0
LCR	27	25	2	92.6	7.4
OIL	74	74	0	100.0	0.0

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Table 2: MutPy Mutation Operators Report

Operator	Total	Non-Eq	Eq	%Non-Eq	%Eq
RIL	26	22	4	84.6	15.4
ROR	196	187	9	95.4	4.6
SCD	2	2	0	100.0	0.0
SCI	1	1	0	100.0	0.0
SDI	233	233	0	100.0	0.0
SDL	818	780	38	95.4	4.6
SIR	7	7	0	100.0	0.0
SVD	652	652	0	100.0	0.0
ZIL	74	74	0	100.0	0.0
Total	3091	3019	72	97.7	2.3

Table 3: Cosmic-Ray Mutation Operators Report

Operator	Total	Non-Eq	Eq	%Non-Eq	%Eq
AddNot	243	243	0	100.0	0.0
ExceptionReplacer	11	11	0	100.0	0.0
NumberReplacer	298	288	10	96.6	3.4
RemoveDecorator	6	6	0	100.0	0.0
ReplaceAndWithOr	22	20	2	90.9	9.1
ReplaceBinaryOperator_Add_BitAnd	17	17	0	100.0	0.0
ReplaceBinaryOperator_Add_BitOr	17	16	1	94.1	5.9
ReplaceBinaryOperator_Add_BitXor	17	17	0	100.0	0.0
ReplaceBinaryOperator_Add_Div	17	17	0	100.0	0.0
ReplaceBinaryOperator_Add_FloorDiv	17	16	1	94.1	5.9
ReplaceBinaryOperator_Add_LShift	17	17	0	100.0	0.0
ReplaceBinaryOperator_Add_Mod	17	17	0	100.0	0.0
ReplaceBinaryOperator_Add_Mul	17	16	1	94.1	5.9
ReplaceBinaryOperator_Add_Pow	17	16	1	94.1	5.9
ReplaceBinaryOperator_Add_RShift	17	17	0	100.0	0.0
ReplaceBinaryOperator_Add_Sub	17	17	0	100.0	0.0
ReplaceBinaryOperator_Div_Add	1	1	0	100.0	0.0
ReplaceBinaryOperator_Div_BitAnd	1	1	0	100.0	0.0
ReplaceBinaryOperator_Div_BitOr	1	1	0	100.0	0.0
ReplaceBinaryOperator_Div_BitXor	1	1	0	100.0	0.0
ReplaceBinaryOperator_Div_FloorDiv	1	1	0	100.0	0.0
ReplaceBinaryOperator_Div_LShift	1	1	0	100.0	0.0
ReplaceBinaryOperator_Div_Mod	1	1	0	100.0	0.0
ReplaceBinaryOperator_Div_Mul	1	1	0	100.0	0.0
ReplaceBinaryOperator_Div_Pow	1	1	0	100.0	0.0
ReplaceBinaryOperator_Div_RShift	1	1	0	100.0	0.0
ReplaceBinaryOperator_Div_Sub	1	1	0	100.0	0.0
ReplaceBinaryOperator_FloorDiv_Add	2	2	0	100.0	0.0
ReplaceBinaryOperator_FloorDiv_BitAnd	2	2	0	100.0	0.0
ReplaceBinaryOperator_FloorDiv_BitOr	2	2	0	100.0	0.0
ReplaceBinaryOperator_FloorDiv_BitXor	2	2	0	100.0	0.0

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Table 3: Cosmic-Ray Mutation Operators Report

Operator	Total	Non-Eq	Eq	%Non-Eq	%Eq
ReplaceBinaryOperator_FloorDiv_Div	2	2	0	100.0	0.0
ReplaceBinaryOperator_FloorDiv_LShift	2	2	0	100.0	0.0
ReplaceBinaryOperator_FloorDiv_Mod	2	2	0	100.0	0.0
ReplaceBinaryOperator_FloorDiv_Mul	2	2	0	100.0	0.0
ReplaceBinaryOperator_FloorDiv_Pow	2	2	0	100.0	0.0
ReplaceBinaryOperator_FloorDiv_RShift	2	2	0	100.0	0.0
ReplaceBinaryOperator_FloorDiv_Sub	2	2	0	100.0	0.0
ReplaceBinaryOperator_Mod_Add	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mod_BitAnd	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mod_BitOr	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mod_BitXor	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mod_Div	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mod_FloorDiv	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mod_LShift	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mod_Mul	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mod_Pow	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mod_RShift	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mod_Sub	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mul_Add	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mul_BitAnd	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mul_BitOr	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mul_BitXor	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mul_Div	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mul_FloorDiv	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mul_LShift	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mul_Mod	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mul_Pow	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mul_RShift	1	1	0	100.0	0.0
ReplaceBinaryOperator_Mul_Sub	1	1	0	100.0	0.0
ReplaceBinaryOperator_Pow_Add	1	1	0	100.0	0.0
ReplaceBinaryOperator_Pow_BitAnd	1	1	0	100.0	0.0
ReplaceBinaryOperator_Pow_BitOr	1	1	0	100.0	0.0
ReplaceBinaryOperator_Pow_BitXor	1	1	0	100.0	0.0
ReplaceBinaryOperator_Pow_Div	1	1	0	100.0	0.0
ReplaceBinaryOperator_Pow_FloorDiv	1	1	0	100.0	0.0
ReplaceBinaryOperator_Pow_LShift	1	1	0	100.0	0.0
ReplaceBinaryOperator_Pow_Mod	1	1	0	100.0	0.0
ReplaceBinaryOperator_Pow_Mul	1	1	0	100.0	0.0
ReplaceBinaryOperator_Pow_RShift	1	1	0	100.0	0.0
ReplaceBinaryOperator_Pow_Sub	1	1	0	100.0	0.0
ReplaceBinaryOperator_Sub_Add	13	12	1	92.3	7.7
ReplaceBinaryOperator_Sub_BitAnd	13	13	0	100.0	0.0
ReplaceBinaryOperator_Sub_BitOr	13	13	0	100.0	0.0
ReplaceBinaryOperator_Sub_BitXor	13	13	0	100.0	0.0
ReplaceBinaryOperator_Sub_Div	13	13	0	100.0	0.0
ReplaceBinaryOperator_Sub_FloorDiv	13	13	0	100.0	0.0
ReplaceBinaryOperator_Sub_LShift	13	13	0	100.0	0.0

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Table 3: Cosmic-Ray Mutation Operators Report

Operator	Total	Non-Eq	Eq	%Non-Eq	%Eq
ReplaceBinaryOperator_Sub_Mod	13	13	0	100.0	0.0
ReplaceBinaryOperator_Sub_Mul	13	12	1	92.3	7.7
ReplaceBinaryOperator_Sub_Pow	13	13	0	100.0	0.0
ReplaceBinaryOperator_Sub_RShift	13	13	0	100.0	0.0
ReplaceBreakWithContinue	2	2	0	100.0	0.0
ReplaceComparisonOperator_Eq_Gt	49	49	0	100.0	0.0
ReplaceComparisonOperator_Eq_GtE	49	48	1	98.0	2.0
ReplaceComparisonOperator_Eq_Is	26	8	18	30.8	69.2
ReplaceComparisonOperator_Eq_IsNot	35	35	0	100.0	0.0
ReplaceComparisonOperator_Eq_Lt	49	48	1	98.0	2.0
ReplaceComparisonOperator_Eq_LtE	49	25	24	51.0	49.0
ReplaceComparisonOperator_Eq_NotEq	49	49	0	100.0	0.0
ReplaceComparisonOperator_GtE_Eq	10	10	0	100.0	0.0
ReplaceComparisonOperator_GtE_Gt	10	10	0	100.0	0.0
ReplaceComparisonOperator_GtE_Is	9	9	0	100.0	0.0
ReplaceComparisonOperator_GtE_IsNot	9	9	0	100.0	0.0
ReplaceComparisonOperator_GtE_Lt	10	10	0	100.0	0.0
ReplaceComparisonOperator_GtE_LtE	10	10	0	100.0	0.0
ReplaceComparisonOperator_GtE_NotEq	10	10	0	100.0	0.0
ReplaceComparisonOperator_Gt_Eq	12	12	0	100.0	0.0
ReplaceComparisonOperator_Gt_GtE	12	9	3	75.0	25.0
ReplaceComparisonOperator_Gt_Is	8	8	0	100.0	0.0
ReplaceComparisonOperator_Gt_IsNot	8	2	6	25.0	75.0
ReplaceComparisonOperator_Gt_Lt	12	12	0	100.0	0.0
ReplaceComparisonOperator_Gt_LtE	12	12	0	100.0	0.0
ReplaceComparisonOperator_Gt_NotEq	12	2	10	16.7	83.3
ReplaceComparisonOperator_IsNot_Eq	4	4	0	100.0	0.0
ReplaceComparisonOperator_IsNot_Gt	4	4	0	100.0	0.0
ReplaceComparisonOperator_IsNot_GtE	4	4	0	100.0	0.0
ReplaceComparisonOperator_IsNot_Is	19	19	0	100.0	0.0
ReplaceComparisonOperator_IsNot_Lt	4	4	0	100.0	0.0
ReplaceComparisonOperator_IsNot_LtE	4	4	0	100.0	0.0
ReplaceComparisonOperator_IsNot_NotEq	4	0	4	0.0	100.0
ReplaceComparisonOperator_Is_Eq	8	0	8	0.0	100.0
ReplaceComparisonOperator_Is_Gt	8	8	0	100.0	0.0
ReplaceComparisonOperator_Is_GtE	8	8	0	100.0	0.0
ReplaceComparisonOperator_Is_IsNot	56	56	0	100.0	0.0
ReplaceComparisonOperator_Is_Lt	8	8	0	100.0	0.0
ReplaceComparisonOperator_Is_LtE	8	8	0	100.0	0.0
ReplaceComparisonOperator_Is_NotEq	8	8	0	100.0	0.0
ReplaceComparisonOperator_LtE_Eq	11	10	1	90.9	9.1
ReplaceComparisonOperator_LtE_Gt	11	11	0	100.0	0.0
ReplaceComparisonOperator_LtE_GtE	11	11	0	100.0	0.0
ReplaceComparisonOperator_LtE_Is	9	8	1	88.9	11.1
ReplaceComparisonOperator_LtE_IsNot	9	9	0	100.0	0.0
ReplaceComparisonOperator_LtE_Lt	11	10	1	90.9	9.1
ReplaceComparisonOperator_LtE_NotEq	11	11	0	100.0	0.0

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Table 3: Cosmic-Ray Mutation Operators Report

Operator	Total	Non-Eq	Eq	%Non-Eq	%Eq
ReplaceComparisonOperator_Lt_Eq	28	28	0	100.0	0.0
ReplaceComparisonOperator_Lt_Gt	28	28	0	100.0	0.0
ReplaceComparisonOperator_Lt_GtE	28	28	0	100.0	0.0
ReplaceComparisonOperator_Lt_Is	15	15	0	100.0	0.0
ReplaceComparisonOperator_Lt_IsNot	15	12	3	80.0	20.0
ReplaceComparisonOperator_Lt_LtE	28	23	5	82.1	17.9
ReplaceComparisonOperator_Lt_NotEq	28	25	3	89.3	10.7
ReplaceComparisonOperator_NotEq_Eq	2	2	0	100.0	0.0
ReplaceComparisonOperator_NotEq_Gt	2	1	1	50.0	50.0
ReplaceComparisonOperator_NotEq_GtE	2	2	0	100.0	0.0
ReplaceComparisonOperator_NotEq_Is	15	15	0	100.0	0.0
ReplaceComparisonOperator_NotEq_IsNot	1	0	1	0.0	100.0
ReplaceComparisonOperator_NotEq_Lt	2	2	0	100.0	0.0
ReplaceComparisonOperator_NotEq_LtE	2	2	0	100.0	0.0
ReplaceFalseWithTrue	23	22	1	95.7	4.3
ReplaceOrWithAnd	7	7	0	100.0	0.0
ReplaceTrueWithFalse	17	17	0	100.0	0.0
ReplaceUnaryOperator_Delete_Not	19	19	0	100.0	0.0
ReplaceUnaryOperator_Delete_USub	3	3	0	100.0	0.0
ReplaceUnaryOperator_USub_Invert	3	3	0	100.0	0.0
ReplaceUnaryOperator_USub_Not	3	3	0	100.0	0.0
ReplaceUnaryOperator_USub_UAdd	3	3	0	100.0	0.0
ZeroIterationForLoop	26	26	0	100.0	0.0
Total	1948	1838	110	94.4	5.6

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